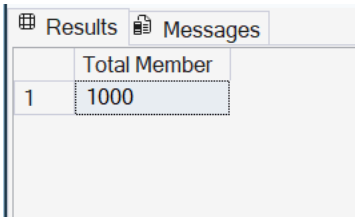
Gym Membership Analysis - SQL Query Report

This project analyzes gym membership, customer behavior, attendance, revenue, and renewal patterns using SQL Server and Power BI. The objective is to help gym management make data-driven decisions to improve customer retention and increase revenue.

- SQL Query Section

1. How many members are registered in the gym?

select count(*) as 'Total Member' from Gym_Data


Results Messages

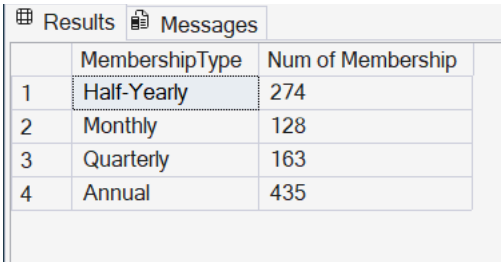
	Total Member
1	1000

2. How many members have Monthly, Quarterly, Half-Yearly, and Annual memberships?

select MembershipType, count(MembershipDuration) as 'Num of Membership'

from Gym_Data

group by MembershipType


Results Messages

	MembershipType	Num of Membership
1	Half-Yearly	274
2	Monthly	128
3	Quarterly	163
4	Annual	435

3. Which membership plan generates the highest revenue?

```
select MembershipType,sum(MembershipFee) as 'Total Revenue'
from Gym_Data
group by MembershipType
order by sum(MembershipFee)  desc
```

Results | Messages

	MembershipType	Total Revenue
1	Annual	6525000
2	Half-Yearly	2192000
3	Quarterly	652000
4	Monthly	192000

4. Which Top 10 membership plan is preferred by each occupation?

```
select top 10 MembershipType,occupation,count(*) as 'Total Member'
from Gym_Data
group by MembershipType,Occupation
order by  count(*) desc
```

Results | Messages

	MembershipType	occupation	Total Member
1	Annual	Freelancer	47
2	Monthly	Student	43
3	Annual	Teacher	40
4	Annual	Civil Engineer	34
5	Annual	Nurse	32
6	Annual	Sales Executive	31
7	Annual	Police Officer	30
8	Annual	Business Owner	29
9	Annual	Accountant	29
10	Half-Yearly	Lawyer	29

5. Which cities generate the highest revenue?

```
select city,sum(membershipfee) as 'Total Revenue'
from Gym_Data
group by city
order by sum(membershipfee) desc ,city
```

Results | Messages

	city	Total Revenue
1	Mumbai	1118500
2	Jaipur	1067500
3	Kolkata	1013000
4	Pune	1000500
5	Patna	965500
6	Lucknow	938000
7	Chandigarh	920500
8	Delhi	898500
9	Hyderabad	874500
10	Bengaluru	764500

6. Which membership plan has the highest renewal rate?

```
SELECT
    MembershipType,
    COUNT(*) AS TotalMembers,
    COUNT(CASE WHEN RenewalStatus = 'Renewed' THEN 1 END) AS RenewedMembers,
    COUNT(CASE WHEN RenewalStatus = 'Renewed' THEN 1 END) * 100.0 / COUNT(*) AS RenewalRate
FROM Gym_Data
GROUP BY MembershipType;
```

Results | Messages

	MembershipType	TotalMembers	RenewedMembers	RenewalRate
1	Half-Yearly	274	191	69.708029197080
2	Monthly	128	41	32.031250000000
3	Quarterly	163	72	44.171779141104
4	Annual	435	344	79.080459770114

7. Who are the most active gym members?

```
SELECT TOP 10
   g.MemberID,
   g.FullName,
   COUNT(a.AttendanceID) AS 'Total Visits'
FROM Gym_Data g
JOIN dbo.Attendence a
ON g.MemberID = a.MemberID
GROUP BY g.MemberID, g.FullName
ORDER BY 'Total Visits' DESC;
```

Results | Messages

	MemberID	FullName	Total Visits
1	192	Ishita Jain	120
2	284	Riya Pandey	120
3	438	Arjun Chauhan	120
4	467	Abhishek Sinha	120
5	367	Divya Yadav	120
6	604	Shreya Yadav	120
7	854	Riya Chauhan	120
8	267	Meera Gupta	120
9	313	Abhishek Joshi	120
10	485	Pooja Sharma	120

8. Which members visit the gym more than the average member?

```
SELECT
   g.MemberID,
   g.FullName,
   COUNT(a.AttendanceID) AS TotalVisits
FROM Gym_Data g
JOIN Attendence a
ON g.MemberID = a.MemberID
```

```
GROUP BY g.MemberID, g.FullName
HAVING COUNT(a.AttendanceID) >
(
   SELECT AVG(VisitCount)
   FROM
   (
      SELECT COUNT(*) AS VisitCount
      FROM Attendence
      GROUP BY MemberID
   ) AS AvgVisit
)
```

Results | Messages

	MemberID	FullName	TotalVisits
1	1	Riya Jain	96
2	6	Arjun Chauhan	91
3	7	Nikhil Jain	114
4	11	Rohan Agarwal	101
5	12	Rohan Jain	92
6	13	Saurabh Das	97
7	15	Divya Joshi	99
8	17	Vikram Joshi	93
9	22	Mohit Sinha	117
10	27	Abhishek Pandey	94
11	28	Arjun Chauhan	117
12	32	Nikhil Singh	105

9. At what hour is the gym busiest?

```
SELECT
   DATEPART(HOUR, CheckIn) AS CheckInHour,
   COUNT(*) AS TotalVisits
FROM Attendence
GROUP BY DATEPART(HOUR, CheckIn)
```

ORDER BY TotalVisits DESC;

	CheckInHour_24hr	TotalVisits
1	17	12997
2	8	12940
3	20	12802
4	18	12796
5	19	12788
6	6	12744
7	7	12593

10. How has revenue changed month by month?

```
SELECT
    YEAR(JoinDate) AS Year,
    MONTH(JoinDate) AS Month,
    SUM(MembershipFee) AS Revenue
FROM Gym_Data
GROUP BY YEAR(JoinDate), MONTH(JoinDate)
ORDER BY Year, Month
```

Results Messages

	Year	Month	Revenue
1	2024	1	338000
2	2024	2	359000
3	2024	3	232500
4	2024	4	319000
5	2024	5	448500
6	2024	6	196000
7	2024	7	287000
8	2024	8	309000
9	2024	9	369000
10	2024	10	417500
11	2024	11	225500
12	2024	12	351000
13	2025	1	291500
14	2025	2	455000
15	2025	3	295500
16	2025	4	522000
17	2025	5	297500
18	2025	6	300500

11. Which cities contribute the most revenue?

```
SELECT
   City,
   SUM(MembershipFee) AS Revenue
FROM Gym_Data
GROUP BY City
ORDER BY Revenue DESC;
```

Results | Messages

	City	Revenue
1	Mumbai	1118500
2	Jaipur	1067500
3	Kolkata	1013000
4	Pune	1000500
5	Patna	965500
6	Lucknow	938000
7	Chandigarh	920500
8	Delhi	898500
9	Hyderabad	874500
10	Bengaluru	764500

12. Which membership type is most popular in each city?

```
WITH MembershipCou AS
(
   SELECT
      City,
      MembershipType,
      COUNT(*) AS TotalMembers,
      ROW_NUMBER() OVER
      (
```

```
        PARTITION BY City
        ORDER BY COUNT(*) DESC
    ) AS RN
  FROM Gym_Data
  GROUP BY City, MembershipType
)

SELECT
  City,
  MembershipType,
  TotalMembers
FROM MembershipCou
WHERE RN = 1;
```

Results | Messages

	City	MembershipType	TotalMembers
1	Bengaluru	Annual	34
2	Chandigarh	Annual	42
3	Delhi	Annual	37
4	Hyderabad	Annual	39
5	Jaipur	Annual	57
6	Kolkata	Annual	49
7	Lucknow	Annual	39
8	Mumbai	Annual	49
9	Patna	Annual	46
10	Pune	Annual	43

13. How does cumulative revenue grow over time?

```
SELECT
  JoinDate,
  SUM(MembershipFee) AS DailyRevenue,
```

```
    SUM(SUM(MembershipFee)) OVER
    (
        ORDER BY JoinDate
    ) AS RunningRevenue
FROM Gym_Data
GROUP BY JoinDate
ORDER BY JoinDate;
```

Results | Messages

	JoinDate	DailyRevenue	RunningRevenue
1	2024-01-01	16500	16500
2	2024-01-02	42000	58500
3	2024-01-03	15000	73500
4	2024-01-06	16500	90000
5	2024-01-07	19000	109000
6	2024-01-09	8000	117000
7	2024-01-10	15000	132000
8	2024-01-11	15000	147000
9	2024-01-12	4000	151000
10	2024-01-13	23000	174000
11	2024-01-14	1500	175500
12	2024-01-15	4000	179500

14. Which age group has the most members?

```
SELECT
    CASE
        WHEN Age BETWEEN 18 AND 25 THEN '18-25'
        WHEN Age BETWEEN 26 AND 35 THEN '26-35'
        WHEN Age BETWEEN 36 AND 45 THEN '36-45'
        ELSE '46+'
    END AS AgeGroup,
    COUNT(*) AS TotalMembers
FROM Gym_Data
```

```
GROUP BY
CASE
    WHEN Age BETWEEN 18 AND 25 THEN '18-25'
    WHEN Age BETWEEN 26 AND 35 THEN '26-35'
    WHEN Age BETWEEN 36 AND 45 THEN '36-45'
    ELSE '46+'
END
ORDER BY TotalMembers DESC;
```

Results | Messages

	AgeGroup	TotalMembers
1	46+	345
2	36-45	237
3	26-35	233
4	18-25	185

15. Which payment mode is used the most?

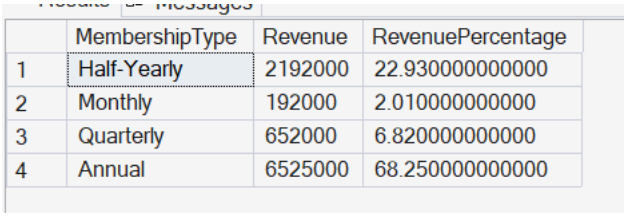
```
SELECT
    PaymentMode,
    COUNT(*) AS TotalPayments
FROM Gym_Data
GROUP BY PaymentMode
ORDER BY TotalPayments DESC;
```

Results | Messages

	PaymentMode	TotalPayments
1	Net Banking	214
2	Credit Card	211
3	UPI	194
4	Cash	192
5	Debit Card	189

16. What percentage of total revenue comes from each membership type?

```
SELECT
   MembershipType,
   SUM(MembershipFee) AS Revenue,
   ROUND(
      SUM(MembershipFee) * 100.0 /
      SUM(SUM(MembershipFee)) OVER (),
      2
   ) AS RevenuePercentage
FROM Gym_Data
GROUP BY MembershipType;
```



	MembershipType	Revenue	RevenuePercentage
1	Half-Yearly	2192000	22.930000000000
2	Monthly	192000	2.010000000000
3	Quarterly	652000	6.820000000000
4	Annual	6525000	68.250000000000

17. Can we create a reusable SQL view for reporting?

```
CREATE VIEW vw_GymDashboard AS
SELECT
   MemberID,
   FullName,
   Gender,
   Age,
```

Occupation,

City,

MembershipType,

MembershipFee,

RenewalStatus,

PaymentMode,

JoinDate

FROM Gym_Data;

To use the view:

SELECT *

FROM vw_GymDashboard;

Results Messages

	MemberID	FullName	Gender	Age	Occupation	City	MembershipType	MembershipFee	RenewalStatus	PaymentMode	JoinDate
1	1	Riya Jain	Female	28	Accountant	Pune	Half-Yearly	8000	Renewed	Debit Card	2025-04-06
2	2	Aditi Das	Female	51	Sales Executive	Kolkata	Quarterly	4000	Renewed	Cash	2026-05-09
3	3	Rohan Verma	Male	26	Nurse	Mumbai	Monthly	1500	Not Renewed	UPI	2025-07-01
4	4	Karan Chauhan	Male	43	Government Employee	Kolkata	Monthly	1500	Not Renewed	UPI	2026-01-01
5	5	Meera Chauhan	Female	38	Civil Engineer	Pune	Half-Yearly	8000	Renewed	Net Banking	2025-04-07
6	6	Arjun Chauhan	Male	52	Sales Executive	Patna	Annual	15000	Not Renewed	UPI	2026-05-28
7	7	Nikhil Jain	Male	57	Doctor	Patna	Half-Yearly	8000	Not Renewed	UPI	2025-06-08
8	8	Abhishek Singh	Male	43	Freelancer	Lucknow	Half-Yearly	8000	Renewed	Debit Card	2024-05-01
9	9	Sakshi Agarwal	Female	54	Banker	Delhi	Half-Yearly	8000	Renewed	Credit Card	2024-05-14
10	10	Aarav Chauhan	Male	25	Freelancer	Hyderabad	Annual	15000	Renewed	UPI	2026-06-04
11	11	Rohan Agarwal	Male	48	Professor	Hyderabad	Half-Yearly	8000	Renewed	Debit Card	2025-01-10
12	12	Rohan Jain	Male	45	Teacher	Kolkata	Annual	15000	Renewed	Cash	2026-06-06

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Overall Findings

The Gym Membership Analysis project was conducted to understand customer behavior, membership trends, attendance patterns, and revenue performance using SQL Server. By analysing the data, several valuable business insights were identified.

Key Findings

- Annual memberships generated the highest revenue and showed the strongest customer retention.
- Monthly memberships had the lowest renewal rate, indicating a higher risk of customer churn.
- Members aged **35+ years** formed the largest customer segment.
- Professionals preferred Half-Yearly and Annual memberships, while students mostly selected Monthly plans.
- Evening hours (around **6 PM–8 PM**) recorded the highest gym attendance, making this the busiest period of the day.
- Digital payment methods such as UPI and cards were used more frequently than cash.
- A small group of members visited the gym significantly more often than the average member, indicating a loyal customer base.
- A few cities contributed a major share of the total revenue, suggesting opportunities for location-based marketing.

Business Recommendations

Based on the SQL analysis, the following recommendations can help improve business performance:

1. Encourage Monthly members to upgrade to Half-Yearly or Annual memberships through discounts and special offers.
2. Introduce a loyalty programme for highly active members by providing reward points, free personal training sessions, or merchandise.
3. Improve retention by sending renewal reminders before membership expiry.
4. Launch referral campaigns to increase new customer acquisition.

5. Schedule additional trainers and equipment availability during peak evening hours to reduce waiting time.
6. Run targeted marketing campaigns in cities with lower membership growth to increase revenue.
7. Offer personalised membership plans based on customer occupation and age group.
8. Promote digital payment options through cashback or discount offers to improve customer convenience.